The University of Azad Jammu and Kashmir

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Course Title: Computer Networks

Course Code: CS-2203

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LAB 02

Introduction to Computer Networks & Cisco Packet Tracer

- ## Introduction:

What is a Computer Network?

A computer network is a digital telecommunications system that allows interconnected nodes to share resources and data. Computing devices communicate with each other using established connections (data links) over cable media (wires, fiber optics) or unwireless media (Wi-Fi).

- ## Network Devices Nodes:

Devices and their functions are given:

- **Router**
  Connects different networks together

- **Switch**
  Connects devices within the same network

- **Firewall**
  Provides network security and controls traffic.

- **End Devices**
  Clients and Servers.

- **Access Point**
  Connects wireless devices to the network.

- ## Methodology

Software used:

Cisco Packet Tracer (Latest Version) - Network simulation software

- ## Lab Procedure

The lab was conducted in four main phases:

1. **Topology Creation**
   Building the network layout

2. **IP Configuration**
   Assigning logical addresses to devices

3. **Connectivity Testing**
   Verifying communication between devices

4. **Command Analysis**

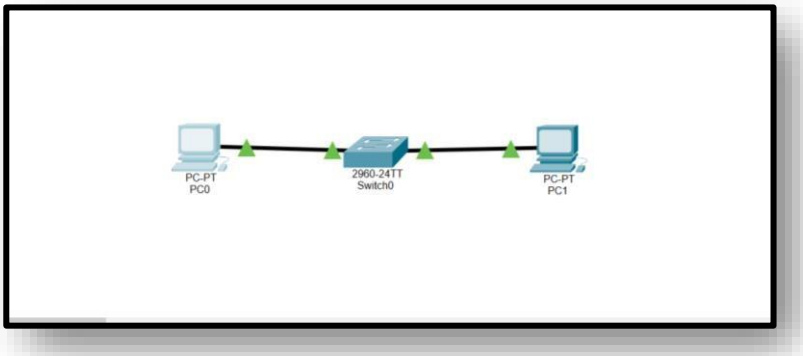
   Using CLI tools to examine network behavior

- ## Lab Task

Part A: Network Topology Creation

Steps:

1. Launched Cisco Packet Tracer
2. Added one Switch (2950-24) from the Network Devices panel
3. Added two PCs from the End Devices panel
4. Connected PC0 to Switch using Copper Straight-Through cable
5. Connected PC1 to Switch using Copper Straight-Through cable



Explanation: The above screenshot displays the complete network topology created in Cisco Packet Tracer. Two end devices (PC0 and PC1) are connected to a central switch (Switch0) using copper straight-through cables. The green indicators on the connection lines confirm successful physical connections between all devices.

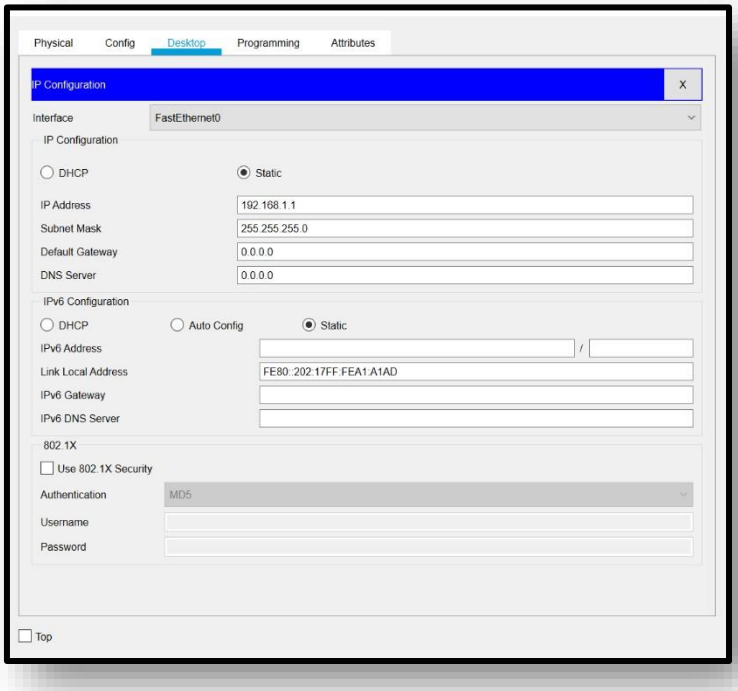
- ## Part B: IP Address Configuration

Following the lab instructions, IP addresses were configured manually on both PCs using the following network parameters:

- **Network Address:** 192.168.1.0
- **Subnet Mask:** 255.255.255.0
- **No Default Gateway required** (no router present)

PC0 Configuration:

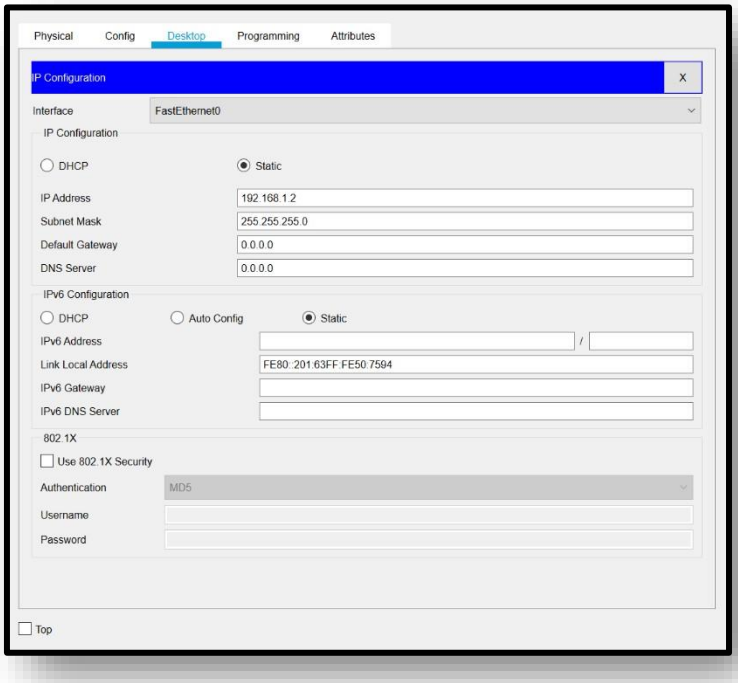
- **IP Address:** 192.168.1.1
- **Subnet Mask:** 255.255.255.0



Explanation: This screenshot shows the IP Configuration window accessed via the Desktop tab on PC0. The configuration is set to "Static" with IP address 192.168.1.1 and subnet mask 255.255.255.0. The Default Gateway field is left blank as no inter-network routing is required for this simple LAN.

PC1 Configuration:

- **IP Address:** 192.168.1.2
- **Subnet Mask:** 255.255.255.0



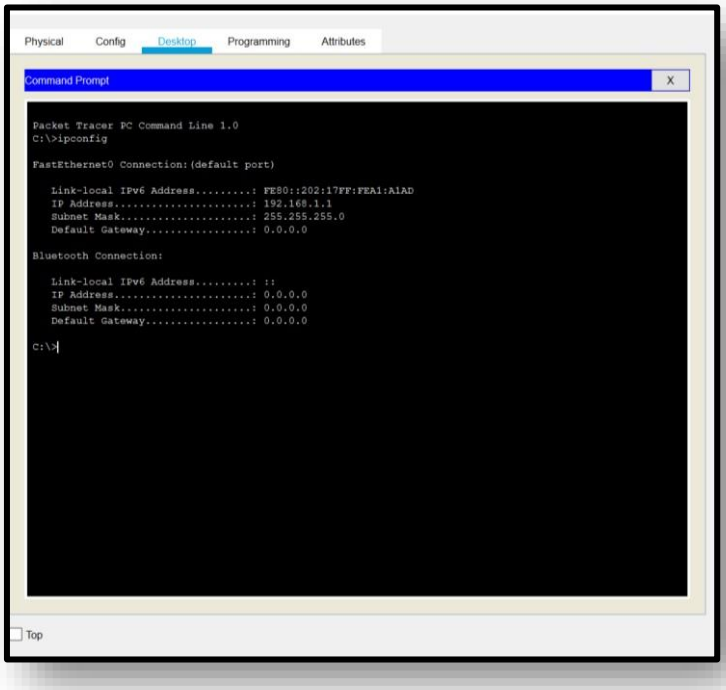
Explanation: PC1 is configured with IP address 192.168.1.2 and the same subnet mask 255.255.255.0. Both PCs now belong to the same logical network (192.168.1.0/24), enabling direct communication between them through the switch.

- ## Part C: Network Testing Using CLI Commands

All testing was performed using the Command Prompt accessed through the Desktop tab on each PC. The following commands were executed and analyzed:

- **Command 1: ipconfig**

Purpose: Displays the current IP configuration of the device.



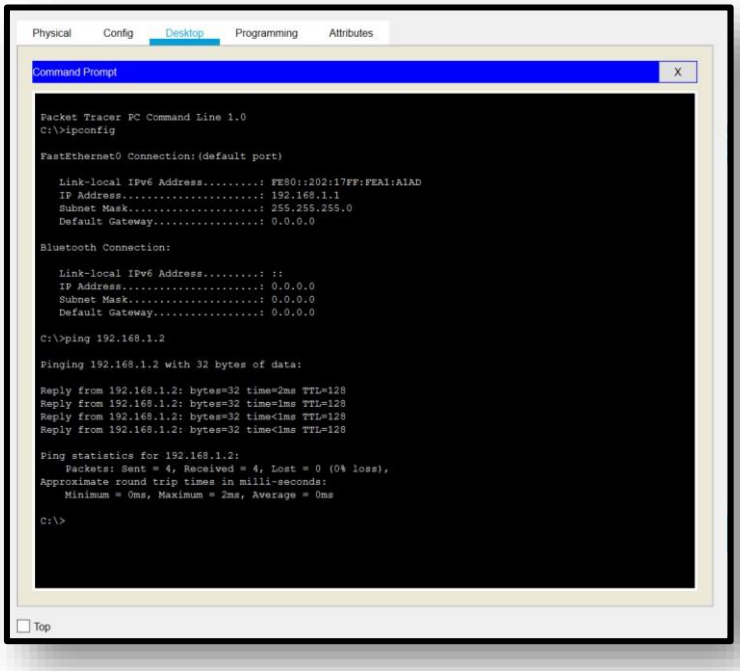
Observation: The output confirmed that PC0 has:

- IPv4 Address: 192.168.1.1
- Subnet Mask: 255.255.255.0
- Default Gateway: (Blank - as expected)

Analysis: The ipconfig command successfully verified that our manual IP configuration was applied correctly. This is the first step in troubleshooting any network issue - confirming that the device has the expected network settings.

- **Command 2: ping**

Purpose: Tests connectivity between two network devices by sending ICMP echo requests.

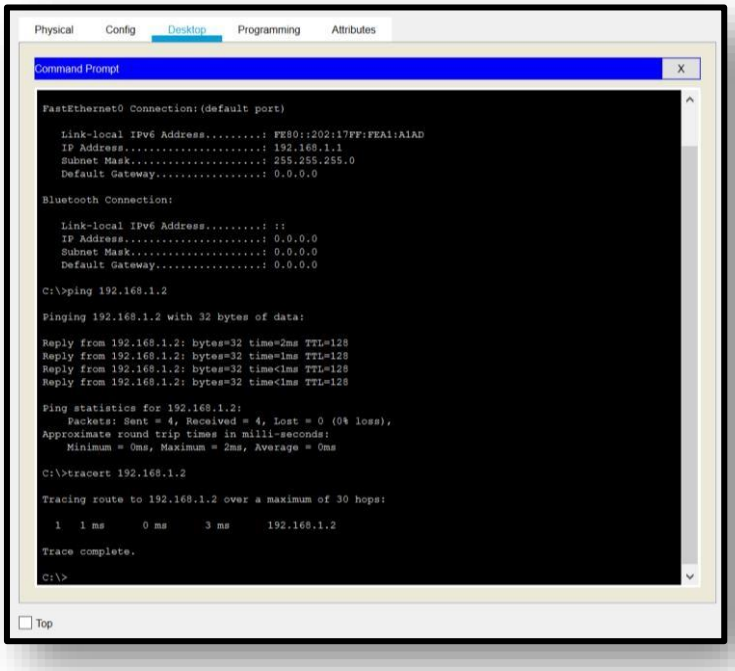


Analysis: The successful ping responses confirm bidirectional connectivity between PC0 and PC1. This verifies that:

1. The physical connections are working correctly
2. IP addresses are configured properly
3. Both devices are on the same network (192.168.1.0/24)
4. The switch is forwarding frames correctly between its ports

- **Command 3: tracert**

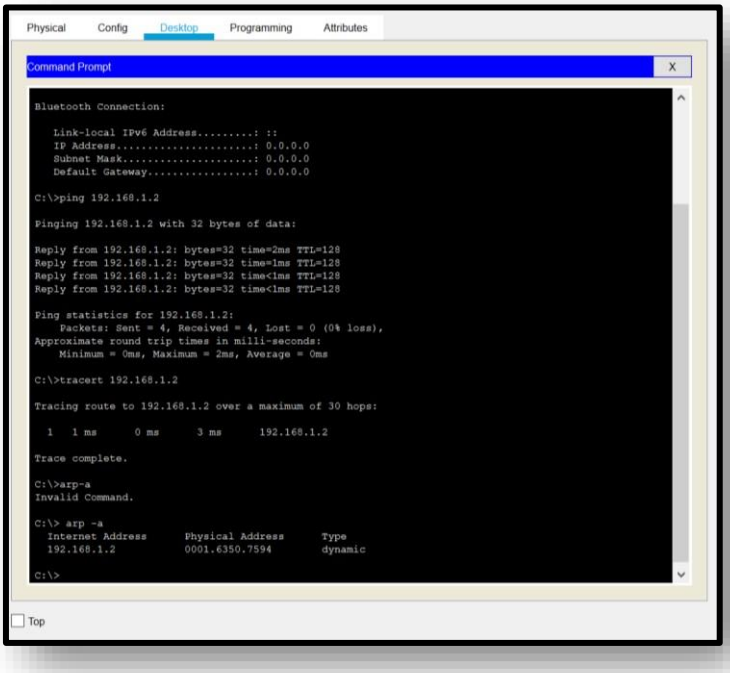
Purpose: Traces the route packets take to reach a destination, showing each hop along the path.



Analysis: The tracert command shows that packets travel directly from the source (PC0) to the destination (PC1) in a single hop. This is expected behavior in a switched network where all devices are on the same subnet. Unlike a router, a switch forwards frames at Layer 2 without decrementing the TTL (Time to Live), so the path appears as a direct connection.

- **Command 4: arp -a**

  Purpose: Displays the ARP (Address Resolution Protocol) cache, which maps IP addresses to physical MAC addresses.



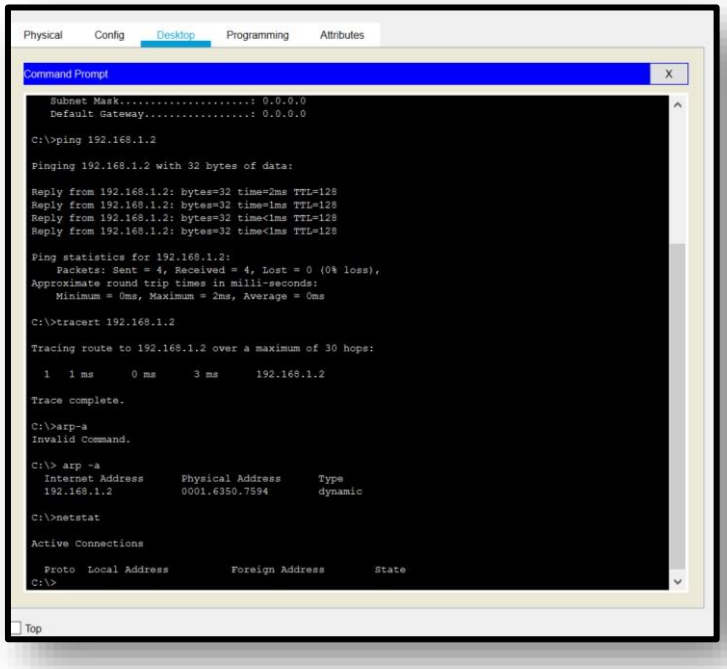
Explanation of ARP: Before PC0 could ping PC1, it needed to know the MAC (Media Access Control) address associated with IP address 192.168.1.2. The ARP protocol works as follows:

1. PC0 broadcasts an ARP request: "Who has IP 192.168.1.2?"
2. PC1 responds with its MAC address
3. PC0 stores this mapping in its ARP cache for future communication

Analysis: After pinging, the arp -a command reveals that PC0 has learned and stored PC1's MAC address. This entry is marked as "dynamic," meaning it will eventually expire and be removed if not used again. This demonstrates how Layer 3 IP addresses are resolved to Layer 2 MAC addresses for actual data transmission.

- **Command 5: netstat**

Purpose: Displays active network connections, listening ports, and network statistics.



Analysis: While the netstat output appears limited in this lab scenario, in real-world networks this command is invaluable for:

- Identifying active connections to remote hosts
- Detecting unauthorized connections
- Troubleshooting applications that aren't communicating properly
- Monitoring network performance and statistics

*****THE END*****